Make NTU

AVNET

03.19.2026

Agenda

Introductions

Meeting goals/objectives

Import SDK

- MCUXPresso Software download
- MCUXpresso IDE
- How to program code

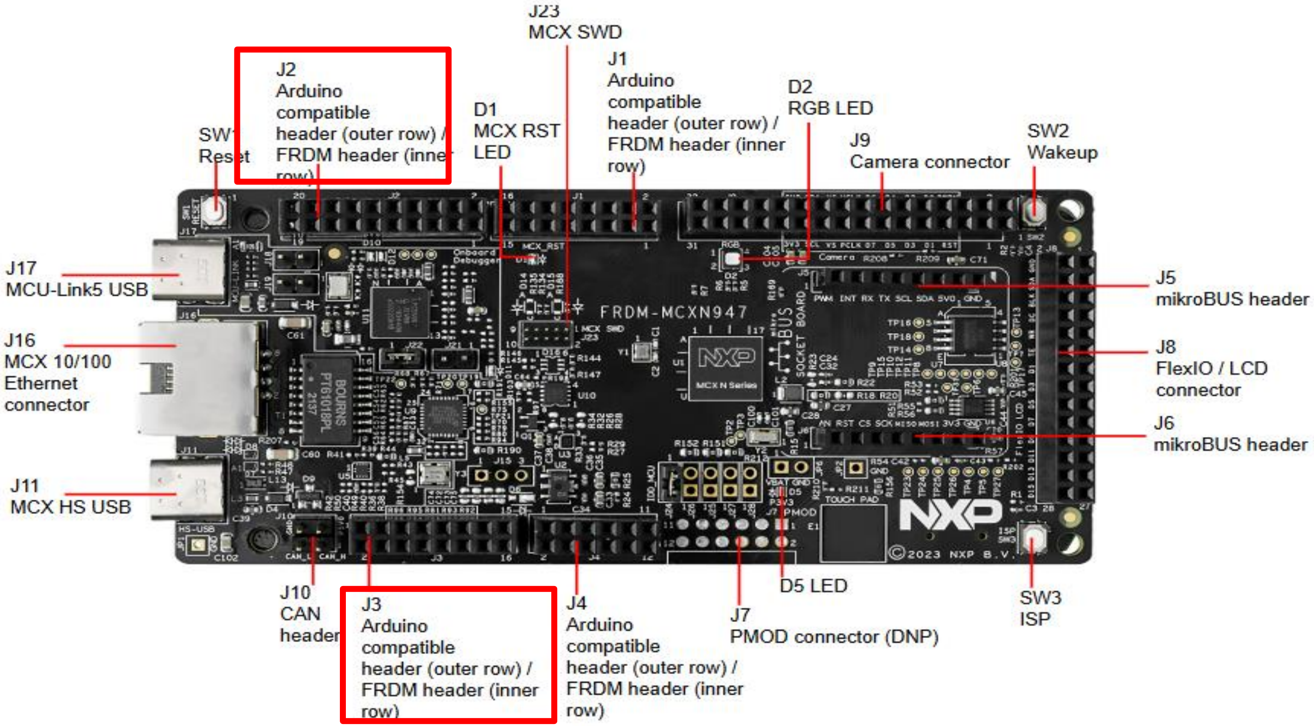
Get To Know MCXN947

NXP Wi-Fi® Modules Using MCXN947 Platform

Q&A

Get To Know MCXN947

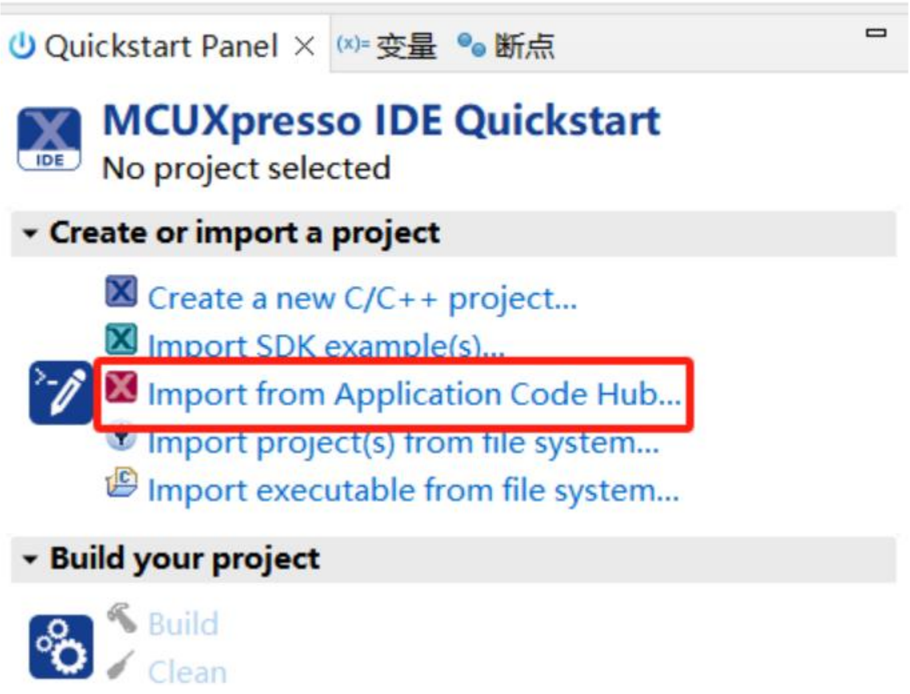
Get Started with the FRDM-MCXN947



FRDM-MCXN947 connectors, push buttons, and LEDs (top-side view)

Import SDK Example

Import SDK Example --- Open MCUXpresso IDE

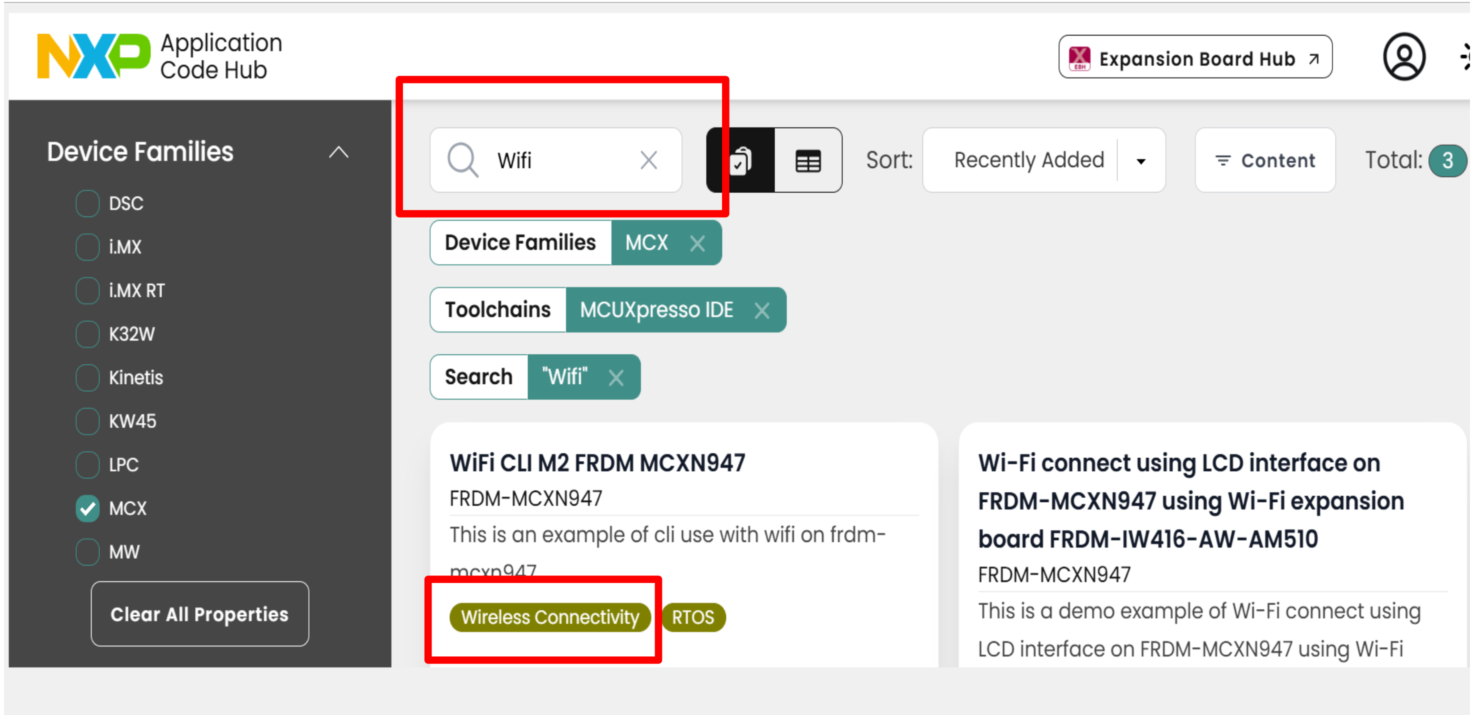


Quick Start Panel, choose Import from Application Code Hub

Import SDK

Import Projects from Application Code Hub

Browse, clone and import application code hub projects. Select a project and click "Copy GitHub link' to continue.



Import SDK example

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Visit on GitHub

GitHub link

Share link

Boards

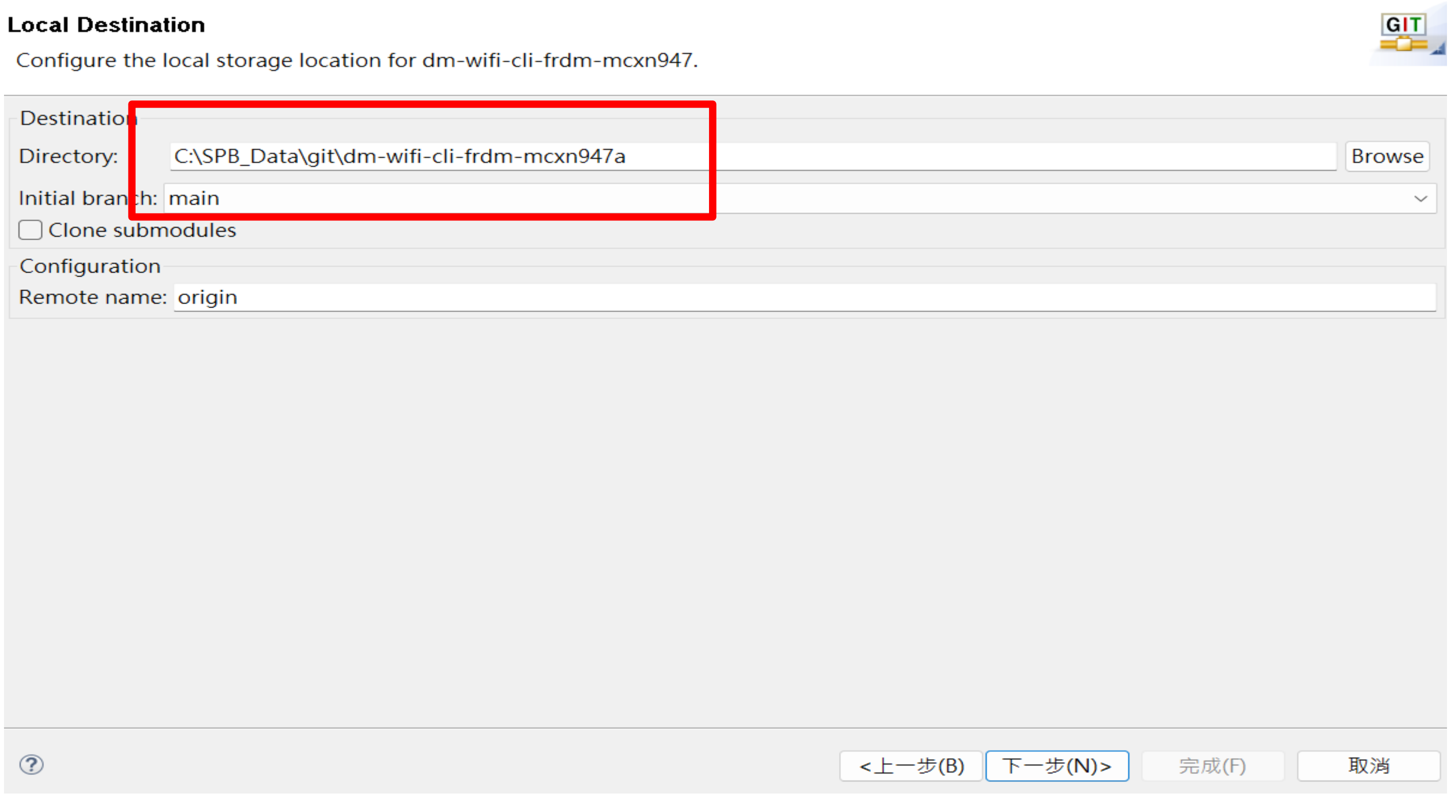
NXP Application Code Hub

WiFi CLI Azure Wave AW-AM510 FRDM MCXN947

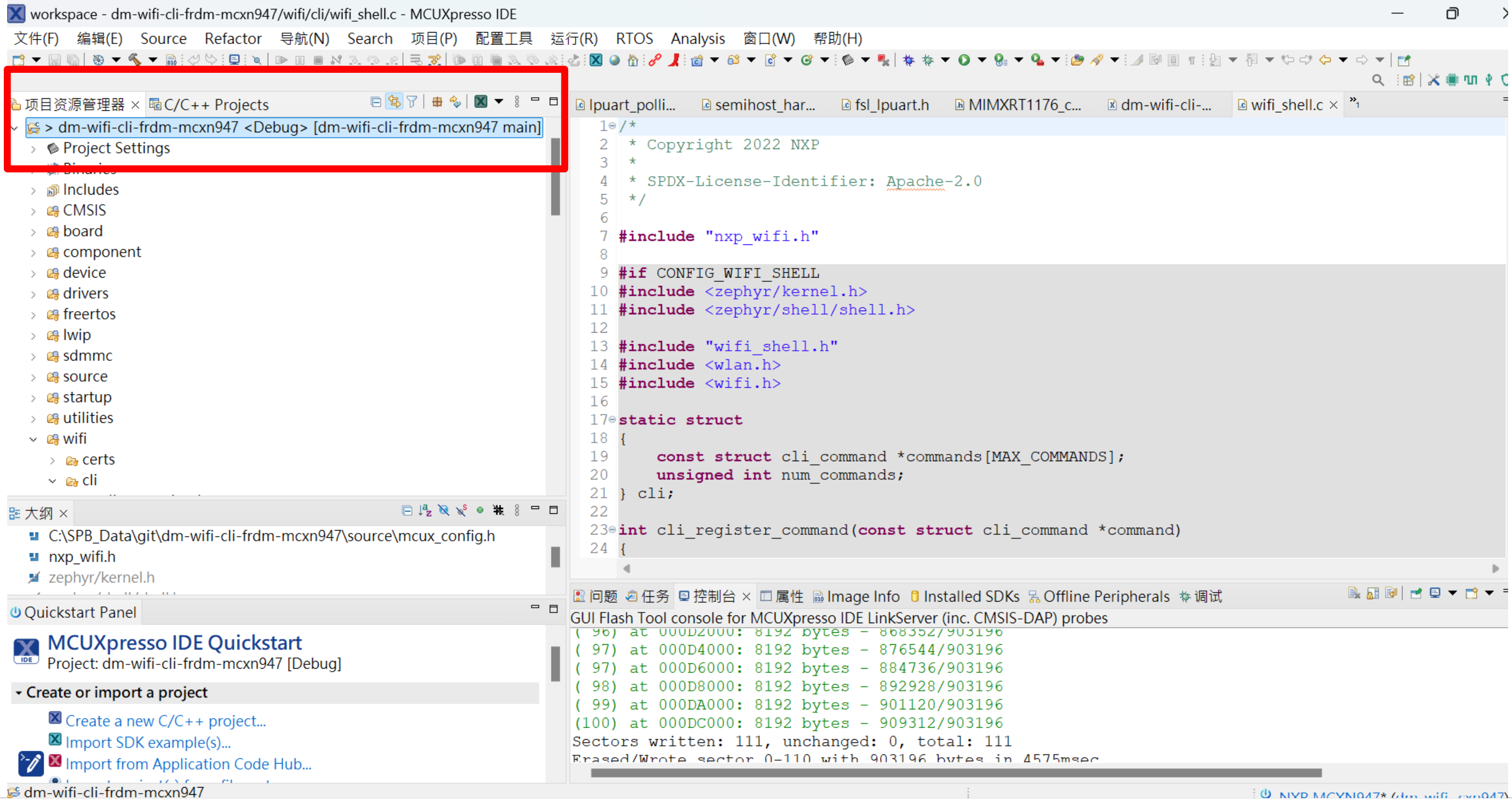
This is an example of cli use with wifi running on serial terminal in frdm-mcxn947, this example has many test options for wifi. There are the options:

```
-help
-wlan-version
```

Import SDK example



Import SDK example



Environment setting and Progrmming

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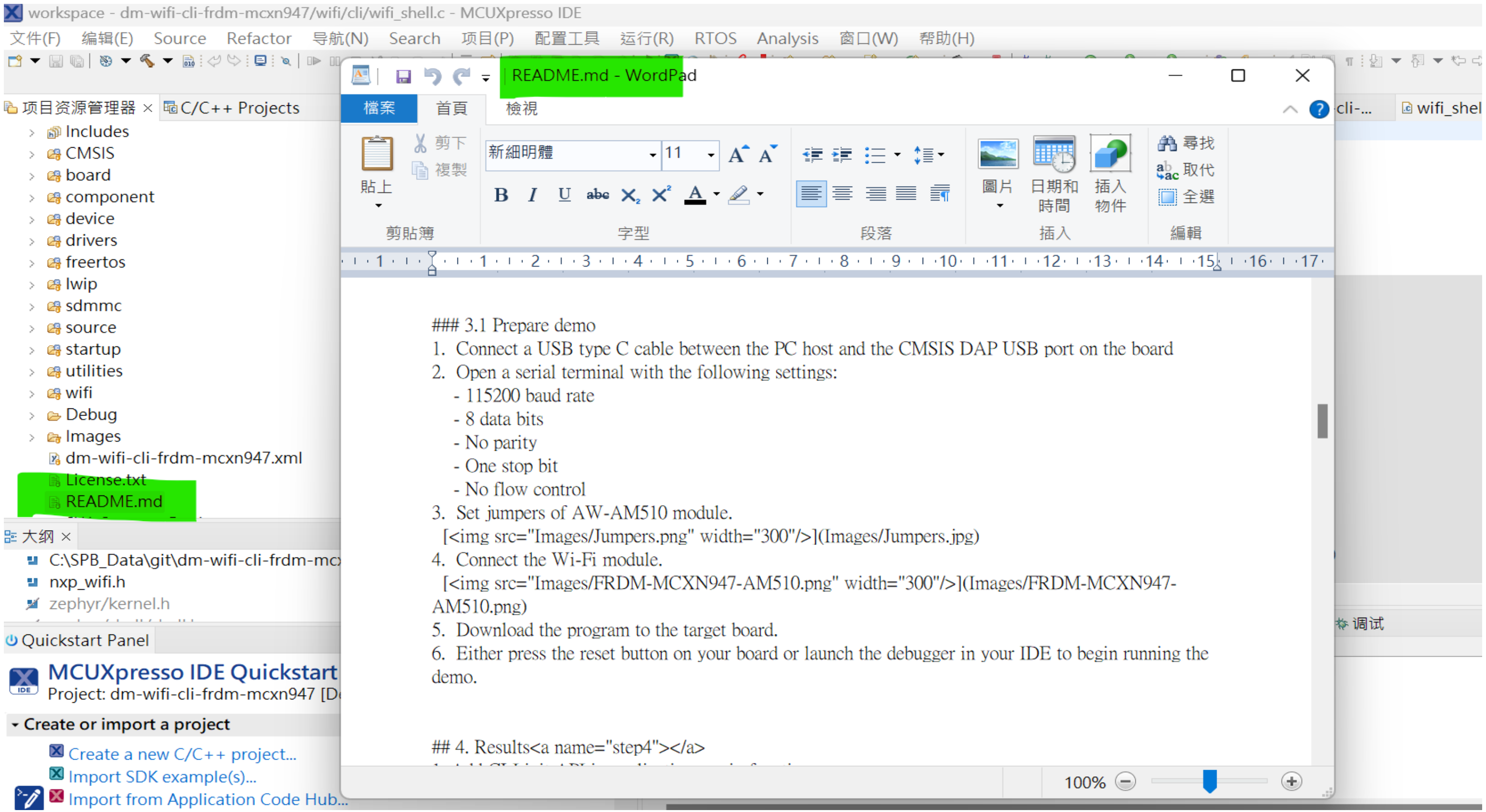
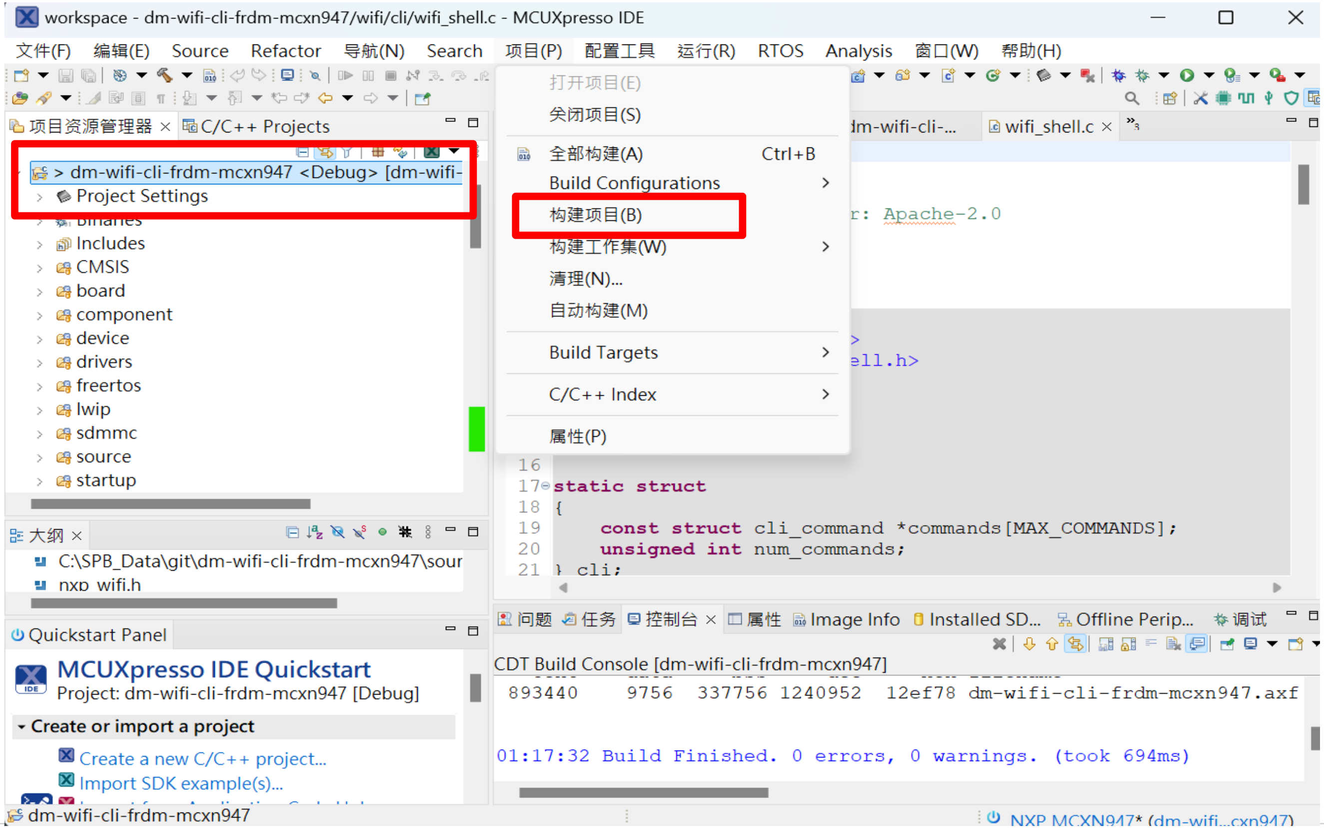
Build and Run Wi-Fi demo from the SDK



How to build

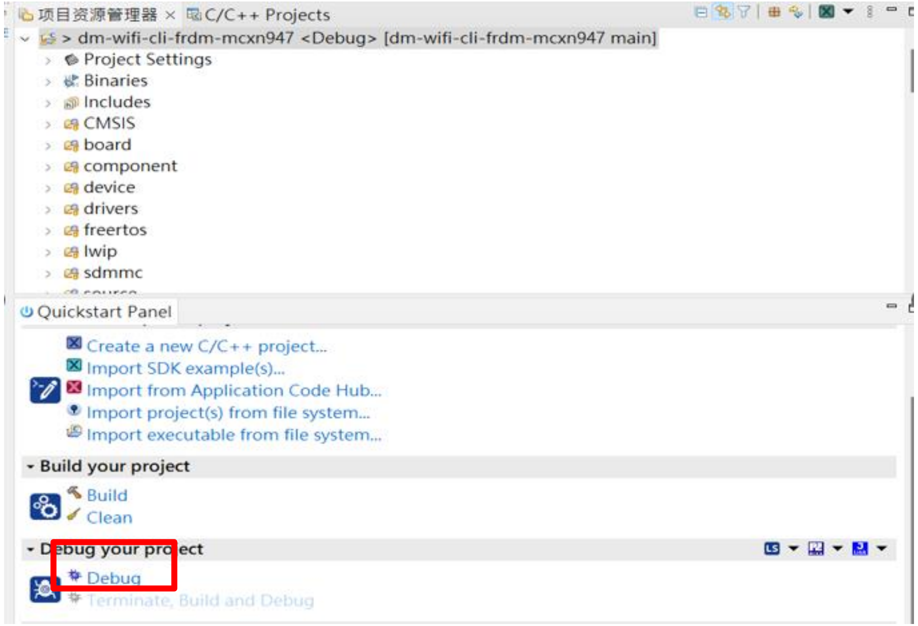
選擇專案右鍵點擊 Build Project, 或是直接點擊



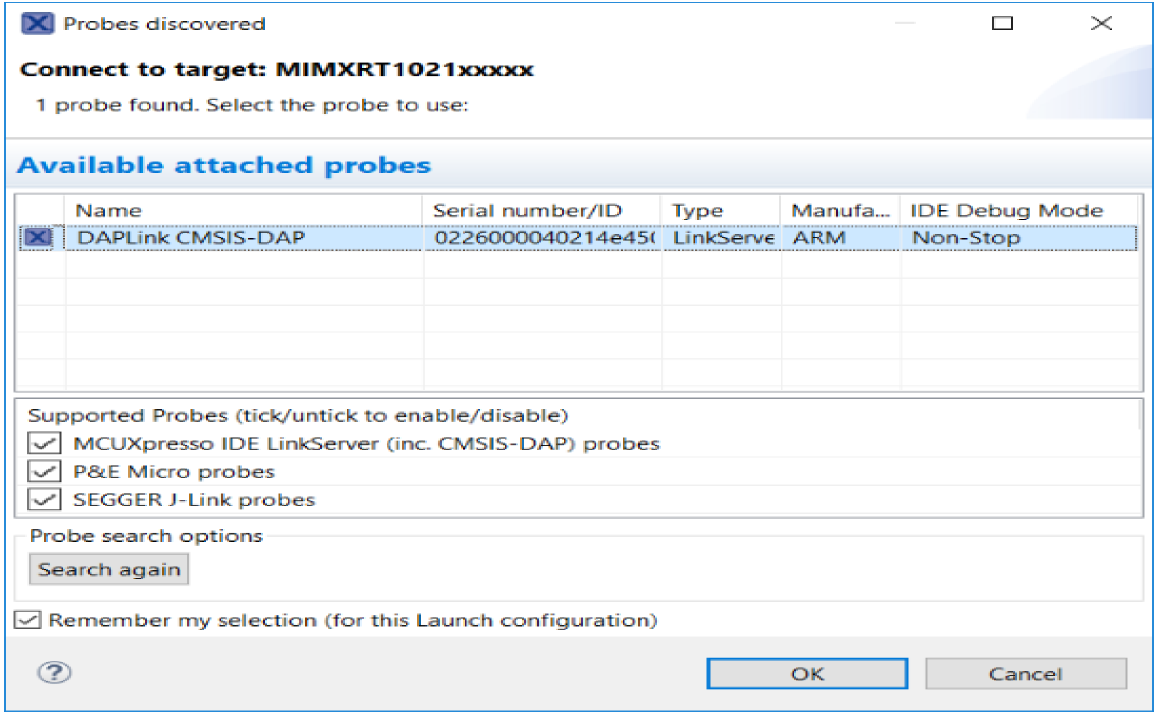
Programming code

1. Now that the project has been complied , you can flash and run it

2. Make sure USB cable plugged into the debug connect on the MCXN947 – EVK , and click on Debug in the QuickStart Panel .



Programming code



Getting Started with NXP Wi-Fi® Modules Using MCXN Platform

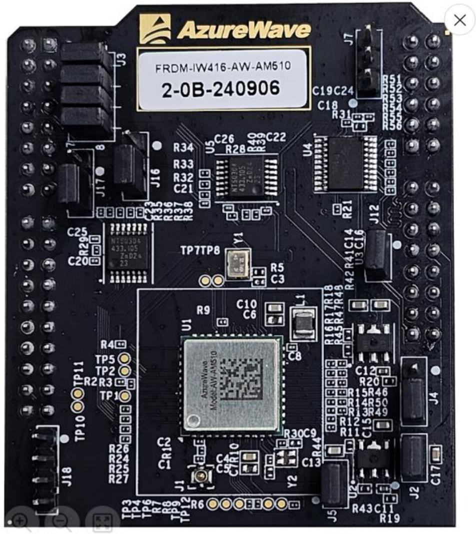
FRDM-IW416-AW-AM510

Wireless Connectivity

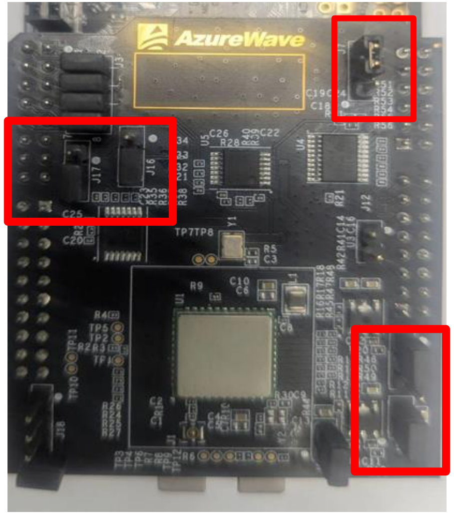
Wi-Fi® + Bluetooth® + 802.15.4

•**IW416**: 2.4/5 GHz Dual-Band 1x1 Wi-Fi® 4 (802.11n) + Bluetooth® 5.2 Solution

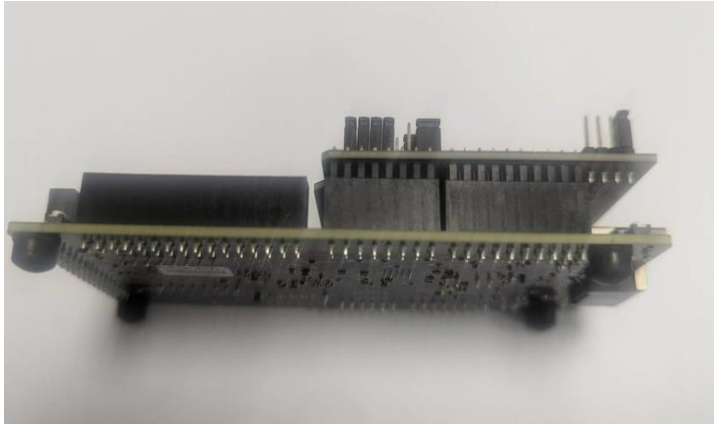
Set jumpers of AW-AM510 module.

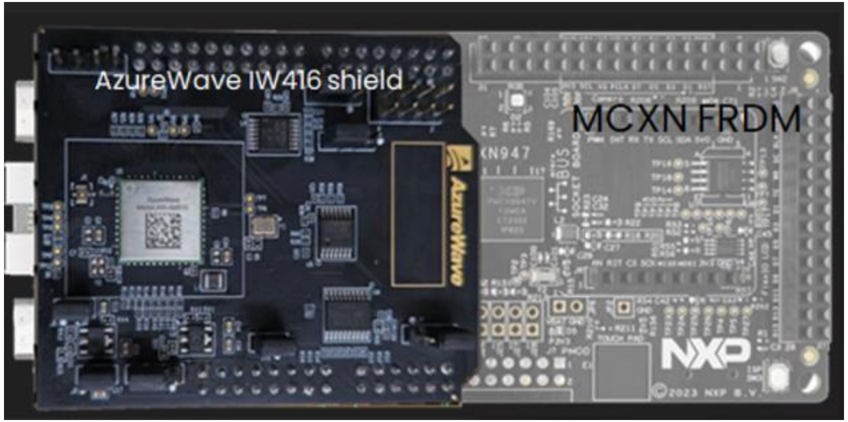


FRDM-IW416-AW-AM510.png



Wi-Fi® Modules Using FRDM-MCXN947 Platform





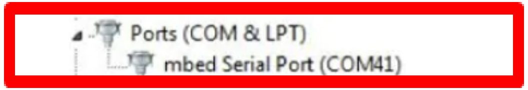
Tera Term

1. **Download** Tera Term from SourceForge. After the download, run the installer and then return to this webpage to continue.

DOWNLOAD

2. **Launch** Tera Term. The first time it launches, it will show you the following dialog. Select the serial option. Assuming your board is plugged in, there should be a COM port automatically populated in the list.

3. **Configure** the serial port settings (using the COM port number identified earlier) to 115200 baud rate, 8 data bits, no parity and 1 stop bit. To do this, go to Setup -> Serial Port and change the settings.



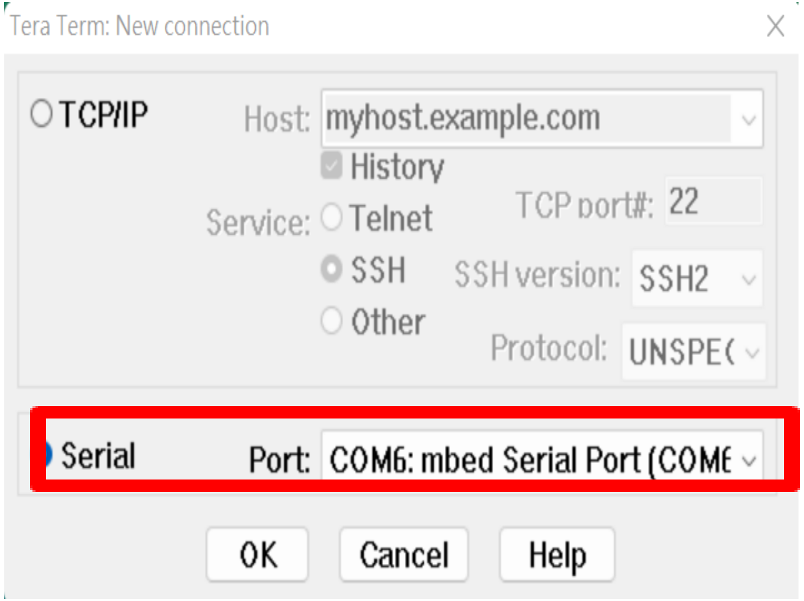
Not sure how to use a terminal application? Try one of these tutorials:

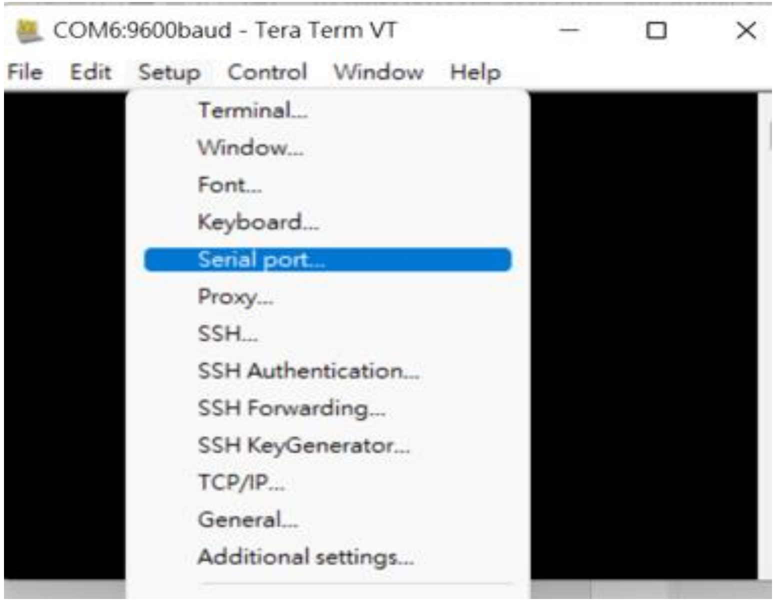
Tera Term Tutorial

PuTTY Tutorial

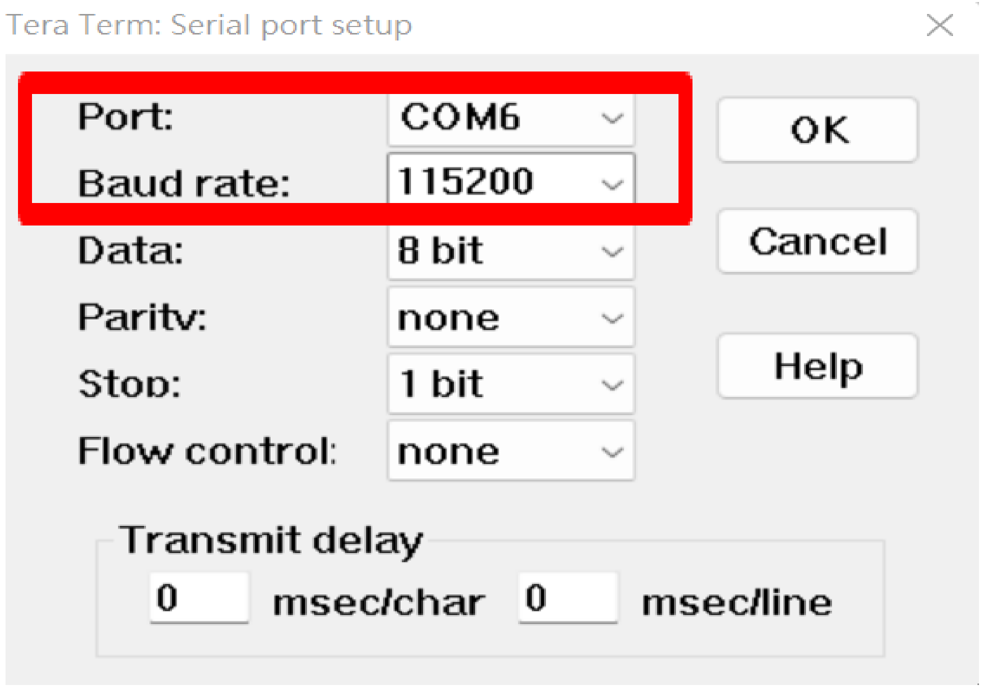
- **Download** Tera Term. After the download, run the installer and then return to this webpage to continue.

Tera Term


Tera Term: New connection
TCP/IP
Host: myhost.example.com
History
Service: Telnet
TCP port#: 22
SSH
SSH version: SSH2
Other
Protocol: UNSPE(
Serial
Port: COM6: mbed Serial Port (COME
OK
Cancel
Help


COM6:9600baud - Tera Term VT
File
Edit
Setup
Control
Window
Help
Terminal...
Window...
Font...
Keyboard...
Serial port...
Proxy...
SSH...
SSH Authentication...
SSH Forwarding...
SSH KeyGenerator...
TCP/IP...
General...
Additional settings...

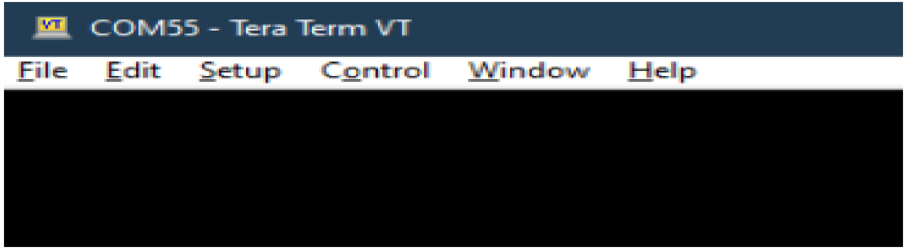
Tera Term



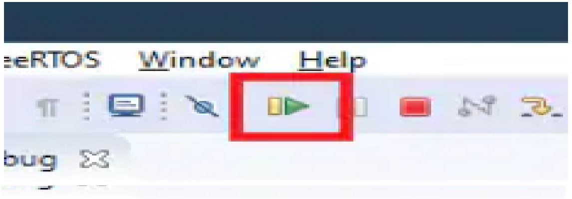
Tera Term: Serial port setup
Port:
COM6
Baud rate:
115200
Data:
8 bit
Parity:
none
Stop:
1 bit
Flow control:
none
OK
Cancel
Help
Transmit delay
0
msec/char
0
msec/line

Building and running a Wi-Fi demo

Before you start your application, open a Serial Terminal application (ie. PuTTY, TeraTerm). You can find instructions to set up a Terminal application on the EVK's Getting Started website → Section 2: Get Software → PC Configuration

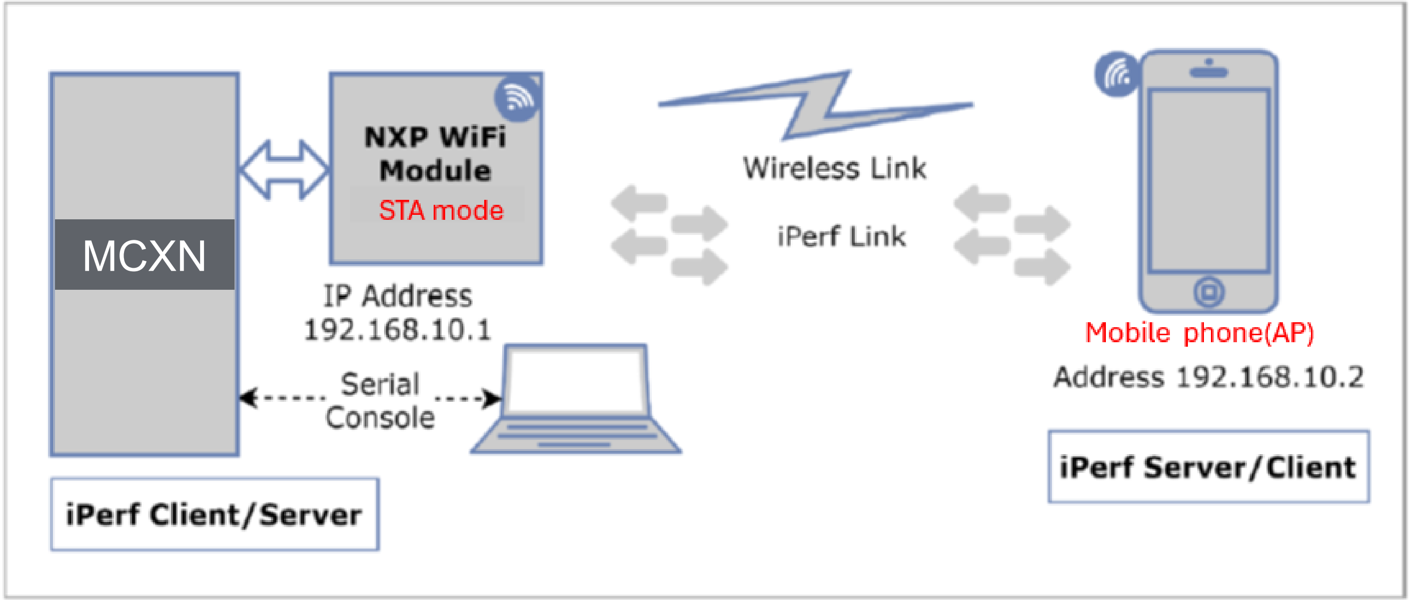


5. Click on the Resume button to start the application

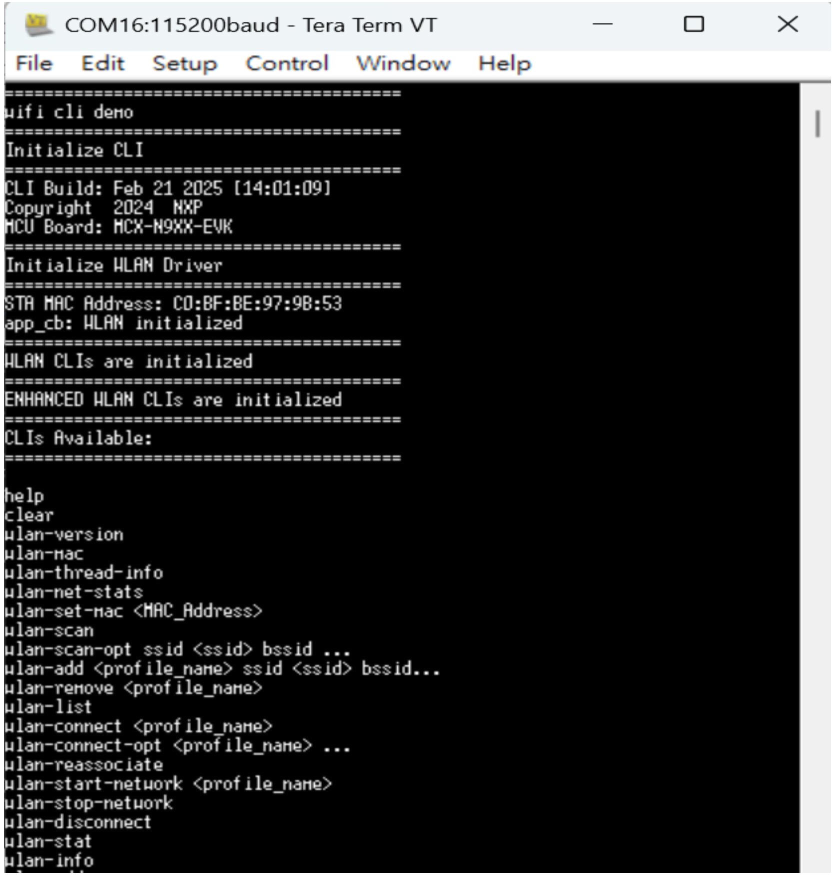


STA/uAP mode

Wi-Fi AP



Running a Wi-Fi demo



Reset Wi-Fi module

Command : wlan-version

```
#
# wlan-version
WLAN Driver Version   : v1.3.r48.p31
WLAN Firmware Version : IW416-V0, RF878X, FP91, 16.92.21.p142.5, WPA2_CVE_FIX 1,
 PVE_FIX 1

# wlan-list
0 networks

# wlan-info
Station not connected
uAP not started

# wlan-address
not connected
```

Scan command # wlan-scan

Command： wlan-scan

```
# wlan-scan
wlan-scan wlan-scan-opt wlan-scan-channel-gap
# wlan-scan
Scan scheduled...

# 2 networks found:
C8:78:7D:68:67:93  "R12-6793" Infra
        mode: 802.11N
        channel: 40
        rssi: -81 dBm
        security: WPA2
        WMM: YES
        802.11K: YES
        802.11V: YES
        802.11W: NO
52:39:97:EF:69:5B  "Byronlink" Infra
        mode: 802.11N
        channel: 48
        rssi: -75 dBm
        security: WPA2
        WMM: YES
        802.11V: YES
        802.11W: Capable
```

此為手機開啟熱點名稱

Add network profile - # wlan-add

```
# wlan-add ?
Usage:
For Station interface
  For DHCP IP Address assignment:
    wlan-add <profile_name> ssid <ssid> [wpa2 <psk/psk-sha256/ft-psk> <secret>]
      If using WPA2 security, set the PMF configuration as mentioned above
    wlan-add <profile_name> ssid <ssid> <owe_only> mfpc 1 mfpr 1
      If using OWE only security, always set the PMF configuration.
      NOTE: [og <"19 20 21">] is only supported in Micro-AP mode .
    wlan-add <profile_name> ssid <ssid> [wpa3 sae <secret> [pwe <0/1/2>] mfpc <1
      If using WPA3 SAE security, always set the PMF configuration.
    wlan-add <profile_name> ssid <ssid> [wpa2 psk psk-sha256 <secret> wpa3 sae <
      If using WPA2/WPA3 Mixed security, set the PMF configuration as mentioned
  For static IP address assignment:
    wlan-add <profile_name> ssid <ssid>
    ip:<ip_addr>,<gateway_ip>,<netmask>
    [bssid <bssid>] [channel <channel number>]
    [wpa2 <psk/psk-sha256/ft-psk> <secret>] [owe_only] [wpa3 sae <secret>] [mfpc
For Micro-AP interface
    wlan-add <profile_name> ssid <ssid>
    ip:<ip_addr>,<gateway_ip>,<netmask>
    role uap [bssid <bssid>]
    [channel <channelnumber>]
    [wpa2 <psk/psk-sha256> <secret>] [wpa3 sae <secret> [pwe <0/1/2>] [tr <0/1>]
    [owe_only ]
    [mfpc <0/1>] [mfpr <0/1>]
```

Command : wlan-add ?

```
# wlan-add aaa ssid Byronlink wpa2 psk 12345678
Added "aaa"
```

Add profile : aaa

Connect to the Station network using the saved network profile:

```
# wlan-conn
wlan-connect wlan-connect-opt
# wlan-connct aaa
command 'wlan-connct' not found

# wlan-conn
wlan-connect wlan-connect-opt
# wlan-connect aaa
Connecting to network...
Use 'wlan-stat' for current connection status.

# app_cb: WLAN: authenticated to network
wapp_cb: WLAN: connected to network
Connected to following BSS:
SSID = [Byronlink]
IPv4 Address: [192.168.30.194]
```

Connect to station

Connect to the Station status:

```
Use 'wlan-stat' for current connection status.

# app_cb: WLAN: authenticated to network
uapp_cb: WLAN: connected to network
Connected to following BSS:
SSID = [Byronlink]
IPv4 Address: [192.168.30.194]


# wlan-st
wlan-start-network wlan-stop-network wlan-stat
# wlan-stat
Station connected (IEEE ps)
uAP stopped
```

Command : wlan-stat

Disonnect to the station status:

```
# wlan-disconnect aaa

# app_cb: disconnected

#
#
# wlan-stat
Station disconnected (Deep sleep)
uAP stopped

#
# wlan-remov

# wlan-remove aaa
Removed "aaa"
```

Command : wlan-disconnect

Command : wlan-remove

Connect to the uAP network using the saved network profile:

```
# wlan-stat
Station disconnected (Deep sleep)
uAP stopped

#
# wlan-add xyz ssid AP ip:192.168.10.1,192.168.10.1,255.255.255.0 role uap wpa2
psk 12345678
Added "xyz"
```

Add uAP profile : xyz

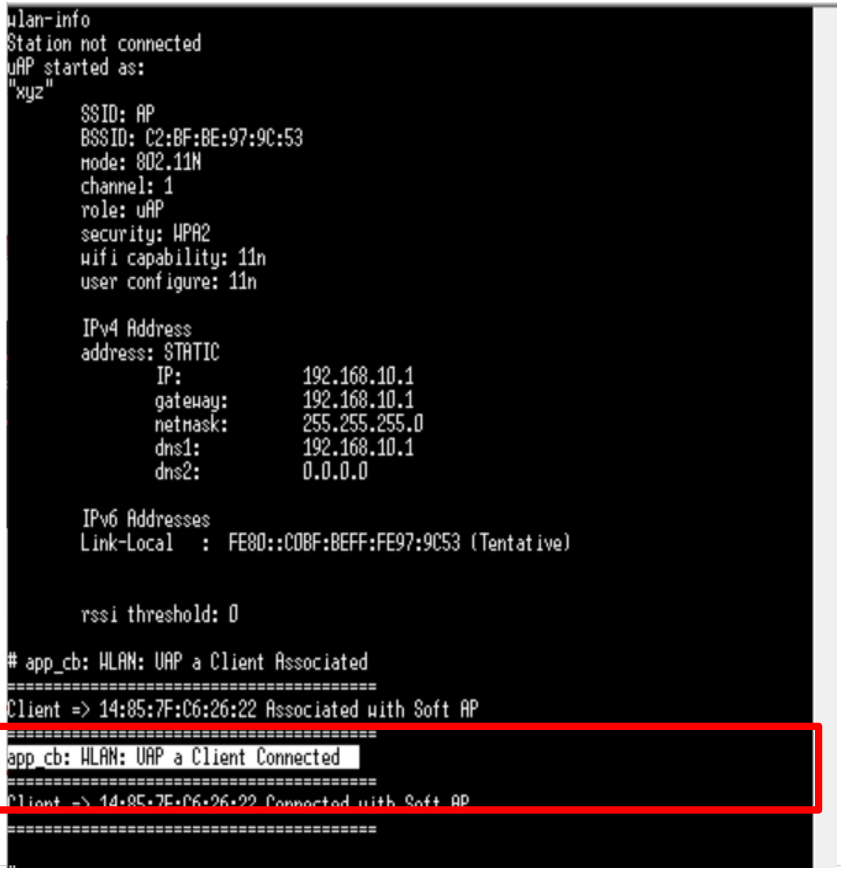
```
# wlan-start-network xyz

# wlaapp_cb: WLAN: UAP Started
=========================================
Soft AP "AP" started successfully
=========================================
nDHCP Server started successfully
=========================================
```

Command : wlan-start-network

```
# wlan-sta
wlan-start-network wlan-stat
# wlan-stat
Station disconnected (Deep sleep)
uAP started (Active)
```

Command : wlan-stat

Connect the uAP From PC:



Ping command format in STA mode

Command : ping [-s <packet_size>] [-c <packet_count>] [-W <timeout in sec>] <ipv4/ipv6 address>

```
ping
Incorrect usage
Usage:
        ping [-s <packet_size>] [-c <packet_count>] [-W <timeout in sec>] <ipv4/ipv6 address>
Default values:
        packet_size: 56
        packet_count: 10
        timeout: 2 sec

# 
```

```
# ping -s 56 -c 2 -W 2 10.163.130.1
PING 10.163.130.1 (10.163.130.1) 56(84) bytes of data
From 10.163.130.194 icmp_seq=1 Destination Host Unreachable
From 10.163.130.194 icmp_seq=2 Destination Host Unreachable

--- 10.163.130.194 ping statistics ---
2 packets transmitted, 0 received, +2 errors, 100% packet loss

# ping -s 56 -c 2 -W 2 10.163.130.194
PING 10.163.130.194 (10.163.130.194) 56(84) bytes of data
64 bytes from 10.163.130.194: icmp_req=1 ttl=255 time=0 ms
64 bytes from 10.163.130.194: icmp_req=2 ttl=255 time=0 ms

--- 10.163.130.194 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss
```

Questions

Thank You

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